

If needful, the reference electrolyte is to be filled, by syringe with needle, after removing the plastic tube from filling hole in the upper part of electrode. After filling the KCl, $c = 0,1 \text{ mol/l}$, the filling hole is covered by plastic tube.

Calibration of the electrode

is made in the range, being expected in samples. A typical calibration curve, over the whole electrode operation range, is supplied with the electrode, and is elaborated especially for this electrode. Calibration standard solutions are prepared from ammonium chloride anhydrous, NH_4Cl . For all standards demineralized water is used. 5,349 g NH_4Cl , dissolved in water and diluted to a volume of 1000 ml gives an ammonium ion solution

- $c = 0,1 \text{ mol/l}$
- $c = 5,349 \text{ g NH}_4\text{Cl/l}$
- $c = 1,400 \text{ g N/l}$
- $c = 1,703 \text{ g NH}_3/\text{l (pH} > 12\text{)}$

The more diluted standards are prepared by exact dilution, together with ionic strength and pH adjustment. The standards of a concentration $0,1 \text{ mmol/l}$ and/or lower are not stable for a long period of time, due to adsorption and microbial consumption. For operating in this range it is recommended to prepare them every day fresh. For calibration standard solutions must be $\text{pH} < 7$, due to pH dependence of $\text{NH}_3 / \text{NH}_4^+$ equilibration, in order to hold all ammonium in the NH_4^+ ionic form, which electrode senses, see equilibration curve. The same is for measured samples. Calibration solutions $c = 0,100 \text{ mol/l}$, $1,00 \text{ gNH}_4^+/\text{l}$ and $1,00 \text{ gN/l}$ are available ready to use.

Distribution of total nitrogen in the $\text{NH}_3 - \text{NH}_4^+$ form, pH dependence

